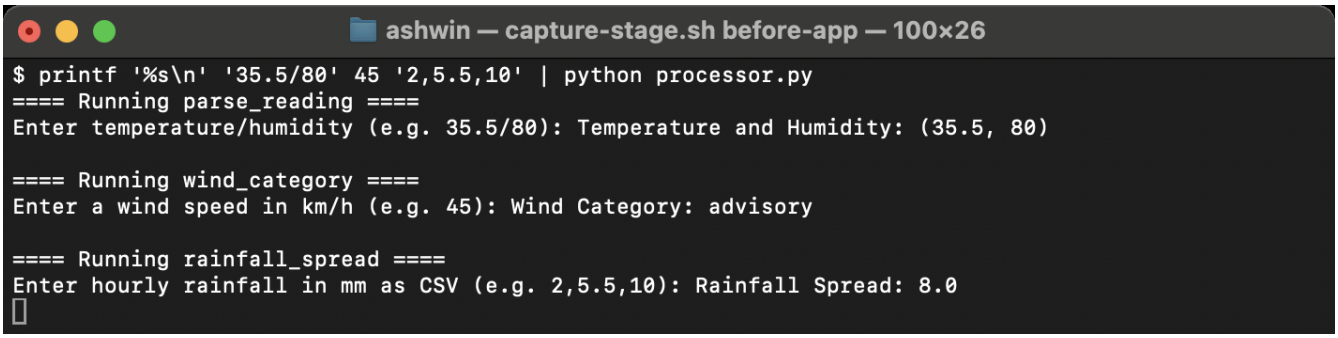
Lab 7 · Fuzz Testing

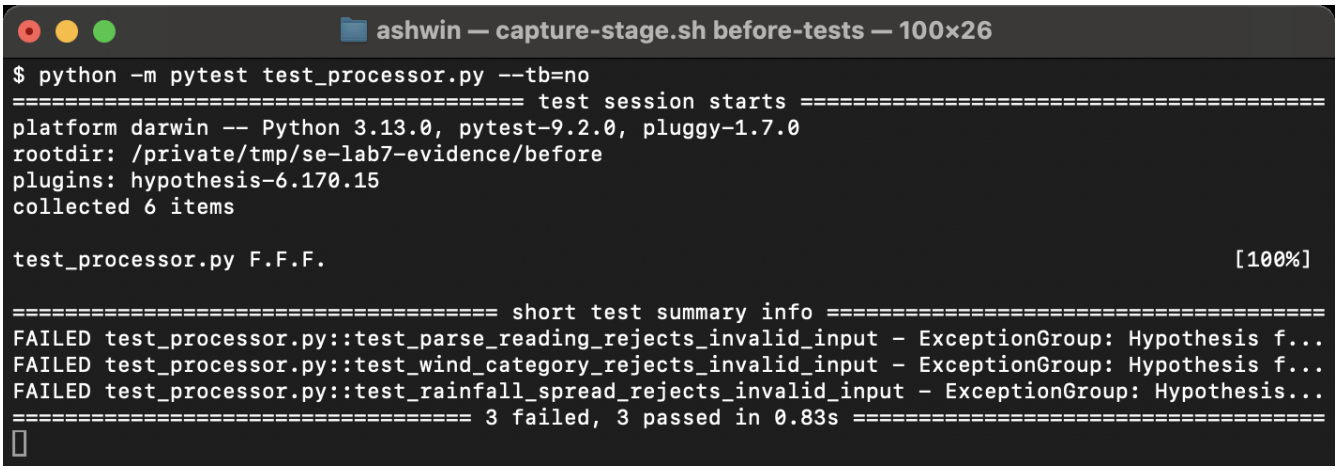
Ashwin Venkatesh | PES1UG24CS090

Hypothesis generated valid and invalid inputs for three weather-processing functions. The starter source came from the assigned weather-alert repository. The submission uses the handout's filenames, **processor.py** and **test_processor.py**.

1. Starter application



2. Fuzz tests against the starter



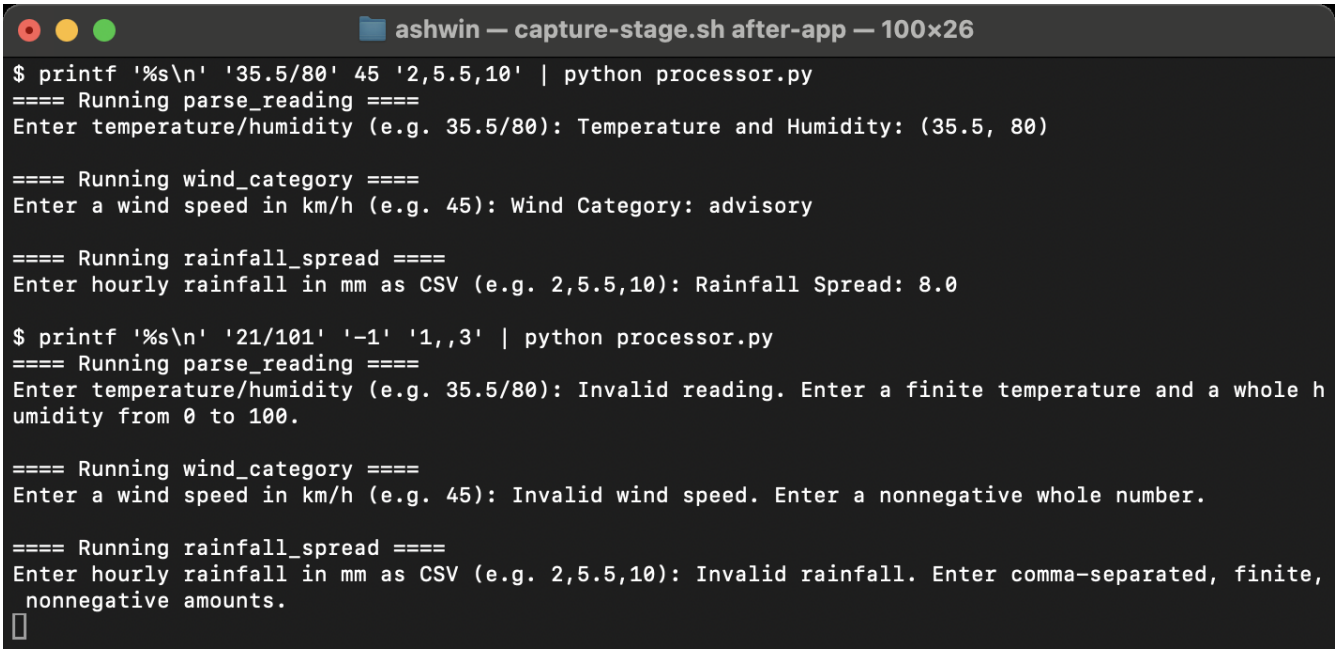
Examples from the failure tracebacks

Function	Generated input	Starter behavior
parse_reading	`"nan/50"`	Accepted a non-finite temperature and returned (NaN, 50).
wind_category	`False`	Coerced the Boolean to zero and returned "none".
rainfall_spread	`"0,nan,3"`	Returned 3.0 despite the invalid middle reading.

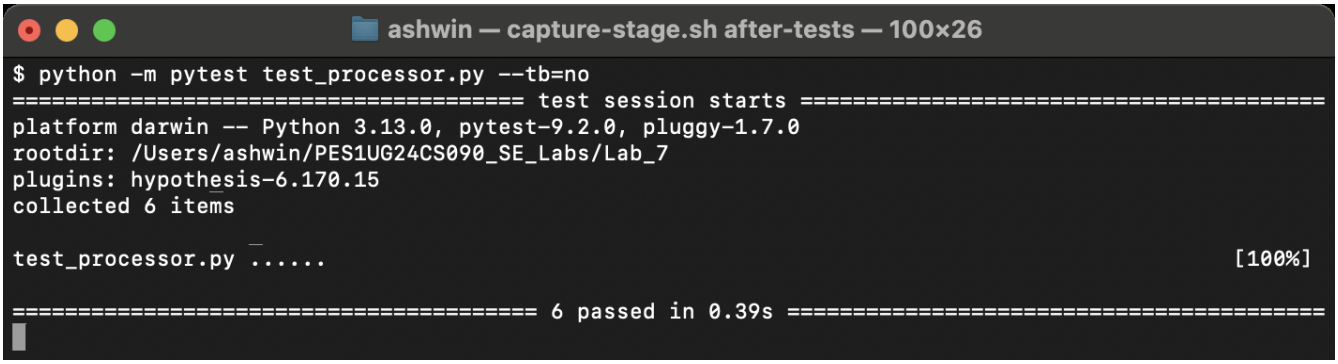
Validation and retest

Each parser returns **None** for invalid input. The reading parser requires a finite temperature and a whole humidity percentage from 0 to 100. Wind speed must be a nonnegative integer, and rainfall must contain at least one finite, nonnegative value. The CLI explains invalid entries and continues to the next prompt.

3. Fixed application



4. Fuzz tests against the fixed application



The same six properties ran in both stages. They verify rejection of invalid inputs, temperature and humidity round-trips, wind thresholds at 39 and 89 km/h, and rainfall spread regardless of reading order.

Reproduce the test run from Lab_7

```
python -m pip install -r requirements.txt
python -m pytest test_processor.py
```

Recorded with Python 3.13.0, pytest 9.2.0, and Hypothesis 6.170.15.